

WHEATON, MN, USA

WMO: 727533

Lat: 45.780N

Lon: 96.545W

Elev: 1025

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04922

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-11.0	-26.1	1.3	-17.3	-18.4	2.1	-10.1	30.3	16.0	27.4	13.7	6.9	280	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	89.8	72.6	86.1	71.3	82.6	69.2	76.5	85.2	74.3	82.5	72.1	80.4	12.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	128.0	82.2	71.9	122.5	80.7	69.8	113.9	78.6	40.6	85.5	38.5	81.2	36.4	79.5	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.9	24.2	20.8	-21.6	94.8	6.8	2.6	-26.5	96.7	-30.5	98.2	-34.2	99.7	-39.2	101.6		
			-21.8	79.7	6.3	2.9	-26.4	81.8	-30.1	83.5	-33.6	85.1	-38.2	87.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	12.0	15.3	28.7	43.6	56.8	66.9	70.8	68.0	60.5	47.0	31.7	17.9
	DBStd	23.16	14.15	13.21	12.98	10.05	8.87	6.50	5.48	5.60	8.34	9.21	11.62	12.30
	HDD50	4822	1179	973	667	235	36	0	0	0	12	167	557	996
	HDD65	8393	1644	1393	1127	642	282	52	14	32	184	562	1000	1461
	CDD50	2415	0	0	5	44	248	509	643	559	326	74	7	0
	CDD65	511	0	0	0	1	28	110	193	127	48	4	0	0
	CDH74	4996	0	0	1	36	394	1079	1731	1134	546	74	1	0
	CDH80	1483	0	0	0	10	127	337	528	303	157	21	0	0
	WSAvg	9.5	9.8	10.1	10.5	11.0	10.9	9.0	7.2	7.1	8.4	9.7	10.0	9.9
	PrecAvg	26.3	0.9	0.7	1.3	2.3	3.0	4.2	3.5	2.7	3.1	2.4	1.0	0.8
Precipitation	PrecMax	32.5	3.0	1.5	3.9	4.5	6.1	8.7	7.4	4.7	7.5	7.0	4.2	2.2
	PrecMin	19.0	0.0	0.1	0.2	0.3	1.2	1.8	1.1	0.6	0.5	0.4	0.0	0.2
	PrecStd	3.8	0.7	0.4	0.8	1.1	1.3	1.9	1.8	1.1	1.8	1.7	1.1	0.6
	PrecStd	3.8	0.7	0.4	0.8	1.1	1.3	1.9	1.8	1.1	1.8	1.7	1.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	50.0	66.1	79.1	90.1	91.4	91.4	90.9	90.3	82.2	66.4	47.7
		MCWB	38.8	44.0	54.8	58.8	66.4	72.7	76.7	75.9	70.0	63.2	54.3	40.0
	2%	DB	38.9	41.5	57.4	71.8	83.5	88.1	88.4	86.4	84.0	73.1	60.6	41.1
		MCWB	35.5	37.0	49.8	53.7	63.8	70.7	75.2	73.4	67.6	57.7	49.3	35.6
	5%	DB	35.8	37.0	51.8	65.6	78.8	83.6	84.5	82.4	81.0	66.4	53.9	36.8
		MCWB	33.1	34.2	45.7	52.2	60.7	68.4	72.7	70.9	66.1	55.6	46.3	33.5
	10%	DB	31.7	34.3	45.3	61.1	72.8	80.8	82.2	80.9	75.4	62.7	47.5	34.2
		MCWB	30.2	32.1	41.1	49.4	58.1	66.2	70.9	69.3	63.7	52.7	41.7	32.1
	0.4%	WB	40.2	44.0	57.2	60.8	69.5	76.4	80.0	79.1	74.2	65.7	56.7	40.8
		MCDB	43.6	48.4	63.2	74.3	81.9	87.3	87.6	87.3	83.5	79.3	64.6	46.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.6	37.6	51.0	56.7	66.2	73.5	77.3	75.5	71.0	60.9	50.4	36.6
		MCDB	39.2	41.0	57.7	68.2	78.2	83.4	86.4	83.1	81.0	70.4	57.7	40.3
	5%	WB	32.8	34.6	45.7	53.2	63.5	70.8	75.1	73.1	68.2	56.7	46.0	34.1
		MCDB	35.1	37.3	50.1	62.8	73.8	80.8	83.0	81.0	76.9	65.3	52.6	36.7
	10%	WB	29.9	32.1	40.8	50.2	60.3	68.0	72.8	70.8	65.8	53.4	42.0	31.8
		MCDB	31.3	34.3	45.8	59.5	70.9	76.6	80.5	78.4	74.3	60.8	47.1	34.2
Mean Daily Temperature Range	5% DB	MDBR	17.0	17.6	16.9	20.1	21.3	19.6	20.0	21.0	22.8	19.5	17.1	15.2
		MCDBR	19.9	19.7	26.1	30.7	29.7	25.4	22.7	23.8	28.4	29.3	26.9	20.2
		MCWBR	17.1	16.3	17.6	16.8	14.6	12.1	12.3	13.1	14.3	16.8	17.7	16.1
	5% WB	MCDBR	18.6	18.6	24.6	26.6	24.6	20.8	20.9	21.7	23.9	25.7	24.9	18.6
		MCWBR	16.3	15.4	17.4	16.2	13.6	12.0	12.7	13.5	14.0	16.0	17.0	15.1
Clear-Sky Solar Irradiance	taub		0.272	0.282	0.311	0.348	0.371	0.385	0.385	0.391	0.357	0.322	0.288	0.267
	taud		2.390	2.387	2.424	2.374	2.380	2.385	2.374	2.342	2.428	2.498	2.506	2.431
	Ebn at Noon		267	287	292	289	284	280	278	272	273	269	261	257
	Edh at Noon		23	28	31	36	37	37	37	37	31	25	20	20
	RadAvg		548	899	1213	1543	1752	1967	2071	1736	1329	840	541	399
All-Sky Solar Radiation	RadStd		58	96	111	116	121	116	82	149	89	112	59	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page